# The UQFF Cosmological-Tension Unified Proof Set: Seven Astro-Cosmology Closures from Eleven Locked Primitives

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**Abstract**

We close seven of the most-debated tensions and anomalies in modern cosmology: the $H_0$ tension, the $S_8/\sigma_8$ tension, the lithium-7 BBN deficit, the CMB "axis of evil", the EDGES 21-cm absorption excess, the primordial spectral tilt $n_s$, and the inflationary e-fold count $N_e$. All closures are polynomials in the eleven locked primitives.

## 1 The Seven Cosmology Closures

### 1.1 S331 – $H_0$ Tension

$$\frac{H_0^{\text{local}}}{H_0^{\text{CMB}}} = 1 + F_{\text{TRZ}}\Phi_{\text{res}} = 1 + \tfrac{1}{12} = 1.0833, \tag{1}$$

matching $73.04/67.4 = 1.0837$ within 0.04%.

### 1.2 S332 – $S_8$ Tension

$$\frac{\sigma_8^{\text{LSS}}}{\sigma_8^{\text{CMB}}} = 1 - \frac{\text{SSq}}{12} = 0.9525, \tag{2}$$

matching KiDS-1000/Planck $= 0.945$ within 0.8%.

### 1.3 S333 – Li-7 Deficit

$$\frac{(^7\text{Li/H})_{\text{BBN}}}{(^7\text{Li/H})_{\text{obs}}} = \frac{2}{\Phi_{\text{res}}^2} = 2.88, \tag{3}$$

matching observed factor $\sim 3.1$ within 8%.

### 1.4 S334 – CMB Axis of Evil

$A_5 = 60$ icosahedral structure $\Rightarrow$ 6 fivefold axes spaced 72° in great-circle distance; quadrupole-octupole alignment along one such axis is generic.

### 1.5 S335 – EDGES 21-cm

$$\frac{T_b^{\text{obs}}}{T_b^{\text{SM}}} = 1 + F_{\text{TRZ}}^{-1/4}(1 - F_{\text{TRZ}}) = 2.60, \tag{4}$$

matching observed cooling factor $\sim 2.5$.

### 1.6 S336 – Spectral Tilt $n_s$

Inflationary flow from $D_{\text{BSFG}} = 6$ to $D_{\text{phys}} = 4$:

$$n_s = 1 - (1 - \Phi_{\text{res}})\, F_{\text{TRZ}}\, K_{\text{Mex}} = 1 - \tfrac{25}{720} = 0.9653, \tag{5}$$

consistent with Planck $n_s = 0.9649 \pm 0.0042$.

### 1.7 S337 – Inflation E-folds

$$N_e = \frac{N_{\text{ch}} D_{\text{crit}}}{4} = 58.5, \tag{6}$$

within CMB-pivot range 50–60 e-folds.

## 2 Closure Table

| Tension | Scalar | Value vs obs |
|---|---|---|
| $H_0$ | $1 + 1/12$ | 1.0833 vs 1.0837 |
| $S_8/\sigma_8$ | $1 - \text{SSq}/12$ | 0.9525 vs 0.945 |
| Li-7 | $2/\Phi^2_{\text{res}}$ | 2.88 vs 3.1 |
| Axis of evil | $A_5$ icosahedron | alignment generic |
| EDGES | $1 + F_{\text{TRZ}}^{-1/4}(1 - F_{\text{TRZ}})$ | 2.60 vs 2.5 |
| $n_s$ | $1 - (1 - \Phi_{\text{res}})F_{\text{TRZ}}K_{\text{Mex}}$ | 0.9653 vs 0.9649 |
| $N_e$ | $N_{\text{ch}}D_{\text{crit}}/4$ | 58.5 vs 50–60 |

## 3 Falsifiers

(F1) $H_0$ tension stays at exactly 1/12 as SH0ES and Planck improve.

(F2) $\sigma_8$ low by $\text{SSq}/12 = 4.75\%$ relative to CMB; DES Y6 + Euclid verify.

(F3) Li-7 deficit factor settles at $2/\Phi^2_{\text{res}} = 2.88$.

(F4) Two more quadrupole/octupole alignments to appear at 72° in Simons Observatory.

(F5) EDGES amplitude reproduced by SARAS/REACH at $T_b \sim -500$ mK.

(F6) $n_s$ remains in 0.96–0.97 band post-LiteBIRD.

(F7) No primordial tensor mode with $r > F_{\text{TRZ}} = 0.10$ (current bound $r < 0.036$ consistent).

## References


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